Pre-Analysis Plan #ManyDaughters

Research Team ID:

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Data and sample

- **Data source**: German Socio-Economic Panel (SOEP), SOEP-Core v39
  - For each hypothesis, we use the most recent survey year in which the corresponding outcome variable was administered: 2016 for Hypotheses 1 and 4, 2019 for Hypothesis 2, and 2021 for Hypothesis 3.
- **Sample**: Parents with at least one biological child. Biological child is selected since sex of child can be chosen when adopting a child.
- **Exclusions**: Childless respondents; incomplete data on key variables.

Stata Code:

```
*** 1. Keep respondents with a biological child ***
keep if biochild_h == 1

*** 2. Generate inDependent variable ***

gen has_daughter = 0
replace has_daughter = 1 if ///
    (pld0175 == 2 | pld0183 == 2 | pld0187 == 2 | pld0191 == 2 | ///
     pld0195 == 2 | pld0199 == 2 | pld0203 == 2)

*** 3. Generate control variables ***

gen num_children = d11107 if d11107$>$0

gen youngest_child_age = pla0003

gen parent_age = syear - bgpbirthy if syear$>$0 & gebjahr$>$0

gen parent_education = pgisced11 if pgisced11$>$0

gen parent_gender = sex

xtile income_quartile = hghinc, nq(4) if hghinc$>$0
```

Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

Variables

Dependent variables Two operationalizations of gender equality attitudes are used:

- `gender_attitude1`: based on item `bgpr384` (attitudes toward equal rights), included if value > 0.
- `gender_attitude2`: first principal component of standardized responses to `bgpr384--bgpr389`, with inversions for negatively worded items (`bgpr387--bgpr389`). All variables are standardized before PCA to account for differing scales.

Main independent variable

- `has_daughter`: binary indicator equal to 1 if the respondent has at least one biological daughter, 0 otherwise.

Control variables

- `num_children`: categorical variable indicating number of children.
- `parent_age`: continuous, age of parent.
- `youngest_child_age`: continuous, age of youngest child.
- `parent_education`: categorical, parent's highest level of education.
- `parent_gender`: binary, 1 = female.
- `income_quartile`: categorical, household income quartile.

Inclusion and exclusion criteria

- Restrict sample to respondents who have at least one biological child.
- Include only respondents from the year 2016 (`year == 2016`).
- Exclude respondents with missing or invalid answers on gender attitude items (`bgpr384--bgpr389` must all be > 0).
- Responses coded with negative values (e.g., "does not apply") are treated as missing (`.`).

Empirical analysis

The following regression model will be estimated:

$$\text{GenderAttitudes}_i = \beta_0 + \beta_1 \cdot \text{HasDaughter}_i + \mathbf{X}_i'\boldsymbol{\beta} + \varepsilon_i$$

Estimation:

- Dependent variable: `gender_attitude1` or `gender_attitude2`
- Main explanatory variable: `has_daughter`
- Controls: `i.num_children`, `c.parent_age`, `c.youngest_child_age`, `i.parent_education`, `i.parent_gender`, `i.income_quartile`
- Estimation method: OLS with robust standard errors

Interpretation of results

- **Test statistic:** We evaluate the t-statistic for β_1 and report the p-value.
- **Decision rule:** A p-value below 0.05 is interpreted as statistically significant at the 5% level (two-sided).
- **Interpretation:** A positive and significant coefficient supports the hypothesis that having a daughter increases support for gender equality. A negative and significant coefficient would contradict the hypothesis. A non-significant result is interpreted as inconclusive.

Stata code

```
* keep relevant year
keep if syear == 2016

*** 1. Generate gender attitudes ***

* Clean variables - replace all negative responses (does not apply,
   ↪ etc) with missing
foreach var of varlist bgpr384 bgpr385 bgpr386 bgpr387 bgpr388
   ↪ bgpr389 {
    replace `var' = . if `var' $<$ 0
}

* A. gender equality defined by same rights
gen gender_attitude1 = bgpr384 if bgpr384 $>$ 0

* B. gender equality as principal components of gender equality
   ↪ attitudes
```

```
keep if bgpr384 $>$ 0 & bgpr385 $>$ 0 & bgpr386 $>$ 0 & bgpr387 $>$
   ↪ 0 & bgpr388 $>$ 0 & bgpr389 $>$ 0

* Invert items with opposite meaning
gen bgpr387_inv = 7 - bgpr387
gen bgpr388_inv = 7 - bgpr388
gen bgpr389_inv = 7 - bgpr389

* Standardize variables
egen bgpr384_std = std(bgpr384)
egen bgpr385_std = std(bgpr385)
egen bgpr386_std = std(bgpr386)
egen bgpr387_inv_std = std(bgpr387_inv)
egen bgpr388_inv_std = std(bgpr388_inv)
egen bgpr389_inv_std = std(bgpr389_inv)

* Principal component analysis
pca bgpr384_std bgpr385_std bgpr386_std bgpr387_inv_std
   ↪ bgpr388_inv_std bgpr389_inv_std

* Generate first principal component
predict pc1, score
rename pc1 gender_attitude2

*** 2. Regression analysis ***

* Regression model
reg gender_attitudeX has_daughter i.num_children c.parent_age c.
   ↪ youngest_child_age ///
    i.parent_education i.parent_gender i.income_quartile, robust
est store tab1

* Save regression table to LaTeX
estout tab1 using "${path}/hypothesis1.tex", style(tex) ///
    cells(b(star fmt(4)) se(par fmt(4))) posthead() ///
    starlevels(* 0.1 ** 0.05 *** 0.01) ///
    stats(r2 N, labels("R2" "Number of observations")) ///
    mlabels(none) collabels(none) eql(none) notype label postfoot( \
       ↪ addlinespace) replace
```

Hypothesis 2: Having daughters leads to more left-leaning political preferences

Variables

Dependent variable

- `left_pref`: binary variable coded as 1 for left-leaning political preferences (`plh0004` between 0 and 4), and 0 otherwise (center or right). Observations with missing or invalid responses are excluded.

Main independent variable

- `has_daughter`: binary indicator equal to 1 if the respondent has at least one biological daughter, 0 otherwise.

Control variables

- `num_children`: categorical variable indicating number of children.
- `parent_age`: continuous, age of parent.
- `youngest_child_age`: continuous, age of youngest child.
- `parent_education`: categorical, parent's highest level of education.
- `parent_gender`: binary, 1 = female.
- `income_quartile`: categorical, household income quartile.

Inclusion and exclusion criteria

- Restrict sample to respondents who have at least one biological child.
- Include only respondents from the year 2019 (`year == 2019`).
- Exclude observations with missing or invalid responses to `plh0004`.

Empirical analysis

The following regression model will be estimated:

$$\text{PoliticalPreferences}_i = \beta_0 + \beta_2 \cdot \text{HasDaughter}_i + \mathbf{X}_i'\boldsymbol{\beta} + \varepsilon_i$$

Estimation:

- Dependent variable: `left_pref`
- Main explanatory variable: `has_daughter`
- Controls: `i.num_children`, `c.parent_age`, `c.youngest_child_age`, `i.parent_education`, `i.parent_gender`, `i.income_quartile`
- Estimation method: OLS with robust standard errors

Interpretation of results

- **Test statistic:** We evaluate the t-statistic for β_2 and report the p-value.
- **Decision rule:** A p-value below 0.05 is interpreted as statistically significant at the 5% level (two-sided).
- **Interpretation:** A positive and significant coefficient supports the hypothesis that having a daughter increases left-leaning political preferences. A negative and significant coefficient would contradict the hypothesis. A non-significant result is interpreted as inconclusive.

Stata code

```
*** 1. Prepare variables ***

* keep relevant year
keep if syear == 2019

* Generate left-leaning political preferences
gen left_pref = .
replace left_pref =  1 if plh0004 $>$= 0 & plh0004 $<$= 4 // left-
   ↪ leaning
replace left_pref =  0 if plh0004 == 5                  // center
replace left_pref =  0 if plh0004 $>$= 6 & plh0004 $<$=10 // right-
   ↪ leaning

*** 2. Regression analysis ***

* Regression model
reg left_pref has_daughter i.num_children c.parent_age c.
   ↪ youngest_child_age ///
    i.parent_education i.parent_gender i.income_quartile, robust
est store tab2

* Save regression table to LaTeX
estout tab2 using "${path}/hypothesis2.tex", style(tex) ///
    cells(b(star fmt(4)) se(par fmt(4))) posthead() ///
    starlevels(* 0.1 ** 0.05 *** 0.01) ///
    stats(r2 N, labels("R2" "Number of observations")) ///
    mlabels(none) collabels(none) eql(none) notype label postfoot( \
       ↪ addlinespace) replace
```

Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)

Variables

Dependent variable

- `male_share_housework`: share of total household work hours completed by the male partner. Calculated as male housework hours divided by total couple housework hours.

Main independent variable

- `has_daughter`: binary indicator equal to 1 if the respondent has at least one biological daughter, 0 otherwise.

Control variables

- `num_children`: categorical variable for number of children.
- `parent_age`: continuous, age of respondent.
- `youngest_child_age`: continuous, age of youngest child.
- `parent_education`: categorical variable for respondent's education level.
- `parent_gender`: binary, 1 = female.
- `income_quartile`: categorical variable indicating income position.

Inclusion and exclusion criteria

- Restrict to individuals living with a partner and with valid reported hours on housework and childcare for both partners.
- Include only observations with valid, positive responses for all relevant hour variables.
- Limit analysis to respondents with at least one biological child.

Empirical analysis

The following regression model will be estimated:

$$\text{HouseholdEqualityIndex}_i = \beta_0 + \beta_3 \cdot \text{HasDaughter}_i + \mathbf{X}_i'\boldsymbol{\beta} + \varepsilon_i$$

Estimation:

- Dependent variable: `male_share_housework`
- Main explanatory variable: `has_daughter`

- Controls: `i.num_children`, `c.parent_age`, `c.youngest_child_age`, `i.parent_education`, `i.parent_gender`, `i.income_quartile`
- Estimation method: OLS with robust standard errors

Interpretation of results

- **Test statistic:** We evaluate the t-statistic for β_3 and report the associated p-value.
- **Decision rule:** A p-value below 0.05 (two-sided) is considered statistically significant.
- **Interpretation:** A positive and significant β_3 implies that having a daughter increases men's share of housework, indicating greater gender equality in the household division of labor. A negative and significant coefficient would suggest reduced equality. A non-significant result is interpreted as inconclusive with respect to the hypothesis.

Stata code

```
*** 1. Prepare variables ***

* keep relevant year
keep if syear == 2021

* Generate weekly hours for childcare and housework
gen hours_childcare_week = .
replace hours_childcare_week = pli0019_h + pli0022_h + pli0044_h if
   ↪ pli0019_h > 0 & pli0022_h > 0 & pli0044_h > 0

gen hours_housework_week = .
replace hours_housework_week = pli0012_h + pli0016_h + pli0043_h if
   ↪ pli0012_h > 0 & pli0016_h > 0 & pli0043_h > 0

* Prepare partner data
preserve
keep pid sex hours_childcare_week hours_housework_week
rename pid parid
rename sex partner_sex
rename hours_childcare_week partner_childcare
rename hours_housework_week partner_housework
tempfile partnerdata
save ‘partnerdata’
restore

* Merge partner data
merge 1:1 parid using ‘partnerdata’
```

```
keep if !missing(hours_childcare_week) & !missing(partner_childcare)
   ↪  & ///
        !missing(hours_housework_week) & !missing(partner_housework)

* Male share of childcare
gen male_childcare = .
replace male_childcare = hours_childcare_week if sex == 1
replace male_childcare = partner_childcare if partner_sex == 1

gen total_childcare = hours_childcare_week + partner_childcare
gen male_share_childcare = male_childcare / total_childcare

* Male share of housework
gen male_housework = .
replace male_housework = hours_housework_week if sex == 1
replace male_housework = partner_housework if partner_sex == 1

gen total_housework = hours_housework_week + partner_housework
gen male_share_housework = male_housework / total_housework

*** 2. Regression analysis ***

* Regression model
reg male_share_housework has_daughter i.num_children c.parent_age c.
   ↪ youngest_child_age ///
    i.parent_education i.parent_gender i.income_quartile, robust
est store tab3

* Save regression table
estout tab3 using "${path}/hypothesis3.tex", style(tex) ///
    cells(b(star fmt(4)) se(par fmt(4))) posthead() ///
    starlevels(* 0.1 ** 0.05 *** 0.01) ///
    stats(r2 N, labels("R2" "Number of observations")) ///
    mlabels(none) collabels(none) eql(none) notype label postfoot( \
       ↪ addlinespace) replace
```

Hypothesis 4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)

Variables

Dependent variable

- `gender_attitude_X`: continuous index measuring attitudes toward gender equality (e.g.,

first principal component of standardized items `bgpr384--bgpr389`).

Main InDependent variables

- `has_daughter`: binary indicator for having at least one biological daughter.
- `female`: binary indicator for being female respondent.
- `female_with_daugther`: interaction term = `has_daughter` × `female`.

Control variables

- `parent_age`: respondent age (continuous).
- `parent_education`: education level (categorical).
- `income_quartile`: income quartile (categorical).
- `num_children`: number of children (categorical).
- `youngest_child_age`: age of youngest child (continuous).

Inclusion and exclusion criteria

- Restrict to respondents with at least one biological child.
- Use pooled data across years with full variable coverage.
- Drop observations with missing values on any variables used in the regression.

Empirical analysis

The following model will be estimated:

$$\text{GenderAttitudeIndex}_{it} = \beta_0 + \beta_4 \cdot \text{HasDaughter}_{it} + \beta_5 \cdot \text{Female}_i + \beta_6 \cdot (\text{HasDaughter}_{it} \times \text{Female}_i) + \mathbf{X}_i' \boldsymbol{\beta} + \varepsilon_i$$

Estimation:

- Dependent variable: `gender_attitude_X`
- Main explanatory variables: `has_daughter`, `female`, and `female_with_daughter`
- Controls: `parent_age`, `parent_education`, `income_quartile`, `num_children`, `youngest_child_age`, `i.syear`
- Estimation method: OLS

Interpretation of results

- **Test statistic:** We focus on the interaction coefficient β_6, and its associated p-value, and report a linear combination test comparing the effect for men and women.
- **Decision rule:** A p-value below 0.05 is considered statistically significant at the 5% level (two-sided).
- **Interpretation:** A significant and negative coefficient on the interaction term (`female_with_daughte`) implies that the effect of having a daughter on gender equality attitudes is larger for men than for women. A non-significant result indicates no statistically meaningful difference across genders.

Stata code

```
*** Run same code as for Hypothesis 2, then continue here ***

gen female = (sex == 2)
gen female_with_daughter = has_daughter * female

reg gender_attitudeX has_daughter female female_with_daughter ///
    i.num_children c.parent_age c.youngest_child_age ///
    i.parent_education i.parent_gender i.income_quartile, robust

_b[female_with_daughter])
```